

Multi-domain topology optimization of connectable lattice structures with tunable transition patterns

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ABSTRACT

The connectivity issue has always been a critical topic in multi-domain topology optimization of lattice structures. In this work, a novel multi-domain topology optimization approach is proposed, in which a set of transitional unit cells that follow a particular varying pattern is introduced between adjacent base microstructures to achieve optimized, multi-class, and well-connected multi-scale structures. Such smooth and tunable transition patterns are realized by interpolation of bar diameter, shape-morphing, and parameterized implicit functions, which exhibit superior adaptability to disconnected microstructures with intricate geometric configurations. To embed the varying transitional unit cells at the specified interface precisely, a graded interface model that extends from the erosion-based interface identification method is established utilizing a linear density filter. Additionally, a SIMP-based multi-domain interpolation scheme considering both base microstructures and transitional unit cells is proposed, in which each piece of interface layer can be further separately defined to navigate situations with more than two lattice material phases. Several 2D and 3D numerical examples are presented to demonstrate the stability, effectiveness, and scalability of the proposed method.

1. Introduction

Inspired by natural structures and materials such as bamboo, bone, and bird beaks, the mechanical behavior of multi-scale structures is significantly influenced by the intricate internal configuration at the microscale and the topology at the macroscale [1–3]. Lattice structures, artificial products derived from natural features, hold the promise for pursuing extraordinary structural performance on a multi-scale owing to their lightweight, high specific stiffness, and compelling multifunctional performance. At the same time, the superior designability characteristic of unit cells allows lattice structures to flexibly customize microstructure configurations in response to varying loading and boundary conditions, which endows them with the ability to navigate complicated stress environments.

Structural topology optimization can be traced back to the pioneering work proposed by Bendsoe and Kikuchi in 1988 [4], in which the numerical homogenization method was implemented to optimize rectangular hole size linked to a periodically distributed

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microstructure. However, the manufacturing limitations at that time enabled this approach to evolve into other single-scale topology optimization methods, e.g., the solid isotropic material interpolation (SIMP) method [5–7], the level set method (LSM) [8,9], the evolutionary structural optimization (ESO) method [10,11], and the moving morphable components (MMC) method [12,13]. Recent advances in additive manufacturing facilitate the fabrication of microstructures with geometric complexity and additionally revive the original homogenization-based topology optimization approach to pursue lightweight structures standing from a multi-scale perspective. In the early work of multi-scale optimization, Rodrigues et al. [14] first proposed a hierarchical topology optimization method to concurrently optimize the local microstructure configurations and the material layout on the macroscale. Nevertheless, since the homogenization method was applied directly without any geometric constraints, the connectivity between neighboring microstructures cannot be guaranteed. To ensure connectivity, Liu et al. [15] assumed that the periodic microstructures are uniform on the macroscale but simultaneously imposed significant limitations on the design space. Based on FE^2 nonlinear multi-scale framework, Xia and Breitkopf [16] developed a concurrent topology optimization to address the nonlinear problem caused by macro–micro adaptation but limited by the intensive computational cost for each material point on macroscale with independent microstructure design.

Extensive research has been done to reach a compromise between design freedom and computational costs. One effective strategy is decomposing the macrostructure into multiple domains, in which cellular microstructures in each domain are assumed to be periodic and uniform at the macroscale. The division of subdomains can follow some heuristic criteria [17–20] or multi-material optimization methods [21]. With a small number of subregions, the design space is expanded to some extent compared with one domain design, and the computational cost is reduced compared with the pointwise design method. Another effective strategy is to construct functionally graded cellular structures. The spatially varying microstructures, featuring similar geometry but varying in size, can be built via one or multiple parameters and allow multi-scale structures to adjust cellular configuration elementwisely within an acceptable computational burden [22–27]. Recently, functionally graded cellular structure optimization methods considering multi-morphology microstructures have received widespread attention [28–32]. Unlike their single-class counterparts, the distribution of domains with multi-class cellular structures and their spatially graded varying are concurrently optimized, which enables them to enlarge the design space further.

Despite many studies focusing on designing multi-scale structures, the connectivity problem remains a challenge since the homogenization-based method is directly applied without considering the behavior around the boundary, especially for dissimilar microstructures. Several strategies have been proposed to pursue connectable multi-scale structures with dissimilar microstructures, further satisfying requirements on manufacturing constraints. The first can be summarized as imposing geometric constraints on base microstructures to generate self-connected multi-scale structures. One representative example is the work reported by Du et al. [33], in which a connectivity index (CI) is proposed as an additional constraint in multi-scale topology optimization to achieve connective microstructures. However, this actually compromises the performance of the optimized design, for it restricts the free-form of the base lattices. Similar works can also be found in [17,34–37], such as adding the kinematical connectors. Another category can be interpreted as performing mandatory connections in the post-processing stage. For instance, constructing the transition microstructures using an interpolation strategy based on bar diameter, morphing, or Sigmoid function after conventional multi-scale topology optimization [32,38,39]; however, such remedies actually ignore the effective performance of the transitional region and possibly deviate from the optimum solution. Considering the above issues, Luo et al. [40] enforced connectivity constraints with a non-design solid domain only on the interface layer, extended from the erosion-based method [41], and coupled the self-connected microstructures into the optimization formula, aiming to eliminate the weakening of the performance of base microstructures and achieve optimal functionally graded cellular structures. Since then, Hu et al. [42] similarly utilized the frame-undesignable domain to generate a connectable lattice structure but built on the two-scale multi-domain concurrent topology optimization framework. Based on that, Gu et al. [43] only reserved the necessary non-design connector on the interface layer and thereby eliminated the redundant connectivity constraints.

Inspired by the work of Luo et al. [40], we focus on multi-domain topology optimization with dissimilar microstructures and present a new strategy to improve connectivity by introducing a set of transitional unit cells that follow a particular transition pattern. The prescribed transition pattern is realized by constructing the graded interface model that enables us to exactly embed the varying transitional unit cells at the specified interface. Different from their work, we target precisely controlling the customizable and smoothly-varying transition unit cells that hold more freedom of shape or topology. Simply adding a layer of non-design domain may disrupt microstructural inter-connectivity and geometry continuity that just endows them with desirable engineering properties, e.g., high mechanical resilience, high energy absorption, and excellent biocompatibility, especially for microstructures with complex and intricate internal configurations such as triply periodic minimal surface (TPMS)-based lattices [44,45]. Furthermore, multiple transition patterns are specified to navigate conditions with more than two lattice material phases, which will involve identifying the interface piece separately. Although the algorithm proposed by Luo et al. [40] is capable of distinguishing interface layers, it has not yet been demonstrated in numerical examples. Therefore, such connectable lattice structures hold the promising potential to alleviate stress concentration induced by abrupt connections between microstructures with different categories and promote other biological functions facilitated by a smoother and gentler transition pattern [46]. Even with complex connectivity problems, especially for lattices featuring significant geometric differences and conditions with multiple dissimilar base microstructures, the proposed method exhibits superior adaptability benefits through the flexibility and customizability of transition patterns achieved by effective interpolation techniques or robust parameterization methods.

As discussed above, establishing a graded interface model is the key to precisely arranging the varied transitional unit cells into the identified interface. In an effort to pursue structures with graded interfaces, LSM becomes the preferred approach due to its advantage in describing structural boundaries. Allaire et al. [47] adopted a signed distance function to interpolate interfacial

properties between different materials smoothly and thereby replaced a discontinuous, sharp interface with a smoothed interface. Since then, Vermaak et al. [48] employed the explicit graded-interface model to research the effect of graded interface on the performance of multi-phase elastic and thermoelastic structures, in which the necessity of consideration of graded interface on topological designs is demonstrated. This model has also been used by Faure et al. [49] and Kang et al. [50] to investigate multi-material periodic microstructures and the robust design of multi-phase structures with uncertain interface zones, respectively. In recent work, Han et al. [51] used the density filter to obtain the position of the element in the interface and mapped it to the graded mechanical properties to solve the stress-based topology optimization problem based on the Bi-directional Evolutionary Structural Optimization (BESO) method. Liu et al. [52] explicitly expressed the graded interface with a controllable width under an element-based topology optimization framework and researched the influence of strong, weak, and intermediate interfaces. However, few studies focus on SIMP-based optimization frameworks, except for the work proposed by Chu et al. [53], in which the almost linearly and definitely monotonically smoothed field was used to construct the graded interface extended from the gradient-based interface identification method [54]. Nevertheless, such an approximate treatment may lead to an inaccurate description of the graded interface and is not applicable to situations with small interface widths.

To address the above challenges, we propose a new strategy to construct graded interfaces under SIMP-based optimization frameworks. Specifically, a linear density filter is adopted to determine the relative position of transition cells in the interface region, further realizing the user-defined transition pattern accurately. The innovation of this work can be stated as follows: (1) a set of transitional unit cells that follow a specific transition pattern are introduced to achieve optimized, multi-class, and well-connected multi-scale structures; (2) each piece of interface layer can be separately defined based on the multi-domain interpolation scheme to navigate situations with more than two lattice material phases; (3) a novel graded-interface model is presented to map smoothly-varying transitional unit cells accurately. The multi-scale structures optimized by the proposed method may exhibit enhanced engineering properties and have adaptability in addressing connectivity issues. On the other hand, the modeled graded interface under SIMP-based optimization frameworks also shows good applicability for situations with small interface widths.

The remaining sections are organized as follows. Section 2 elaborates on the multi-domain topology optimization algorithm for connectable lattice structures, which involves the three-step graded interface identification method and interpolation scheme for multi-domain connectable lattice structures. In Section 3, the optimization formulation with the objective of compliance minimization under volume constraint is mathematically proposed, and the sensitivity is analytically derived. In Section 4, we employ several 2D and 3D numerical examples to demonstrate the stability, effectiveness, and scalability of the proposed method. Finally, some concluding remarks are given.

2. Model of multi-domain connectable lattice structures

2.1. General idea

If no geometric constraints are imposed on the unit cell boundaries between microstructures with dissimilar topologies, a disconnected or abrupt interface will be generated, as shown in Fig. 1. To achieve well-connected lattice structures and facilitate an effective manufacturing process, we introduce a set of transitional unit cells that enable smooth connections between adjacent microstructures. The proposed topology optimization methodology is summarized in a diagram (Fig. 2). We first define the transition unit cells that follow a particular transition pattern through bar diameter or shape-morphing interpolation or are adjusted more delicately to address practical manufacturing resolution requirements. In this way, the candidate base microstructures break free from self-connected restrictions and gain more topological freedom due to the customizable transition pattern. Even microstructures featuring significant differences in configuration can be well-connected with the help of the robust parameterization method, given a sufficient number of transition units.

Then, we collect offline microstructure data involving relative density and equivalent elastic tensors to perform multi-domain topology optimization of connectable lattice structures later. Taking cross and x-shape microstructures as an example (Fig. 2), nine hybrid microstructures are collected by bar diameter interpolation. Note that the actual transitional unit cells may not precisely be the underlying collected microstructures but are determined by the variables for controlling the graded interface and the parameters related to the interface width. The indices of collected microstructures, i.e., the horizontal coordinates of discrete points in Fig. 2, are normalized to range of 0–1 with equal intervals, aiming to map graded variables. Aside from these discrete points, linear interpolation is used to establish the mapping relationships between graded variables, relative density, and equivalent elastic tensors of all underlying lattices. Here, the equivalent elastic tensors of each microstructure are obtained by the homogenization method, which can be implemented in MATLAB and referred to [55,56].

We define the matrix of design variables, $\mathbf{X} = \{x_{e,1}, \dots, x_{e,M}\}_1^N$, to find the optimal design composed of $M + 1$ base domains and their separate interface layers. N is the number of elements that are used to discretize the design domain Ω . Based on each $x_m, m = 1, \dots, M$, we adopt the erosion-based method [41], which performs two-step filtering and two-step projection to derive the intermediate variables μ_m and the eroded variables ϕ_m successively, such that the nonintersecting partitions are mathematically represented based on the SIMP multi-material interpolation scheme [57,58]. Finally, we introduce the smoothed variables $\gamma_m, m = 1, \dots, M$, to generate a graded interface layer along with the thickness direction, aiming to interpolate in the offline databases and embed the varying transitional unit cells precisely.

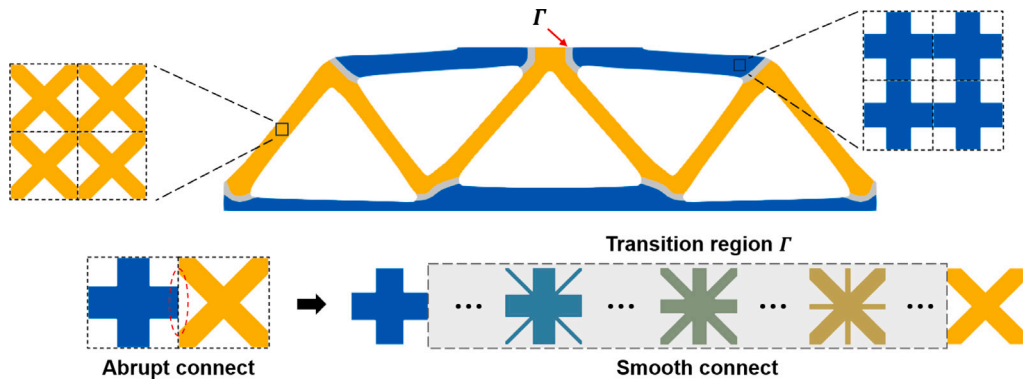


Fig. 1. The abrupt connection between dissimilar adjacent microstructures, and smooth connection by introducing transitional unit cells.

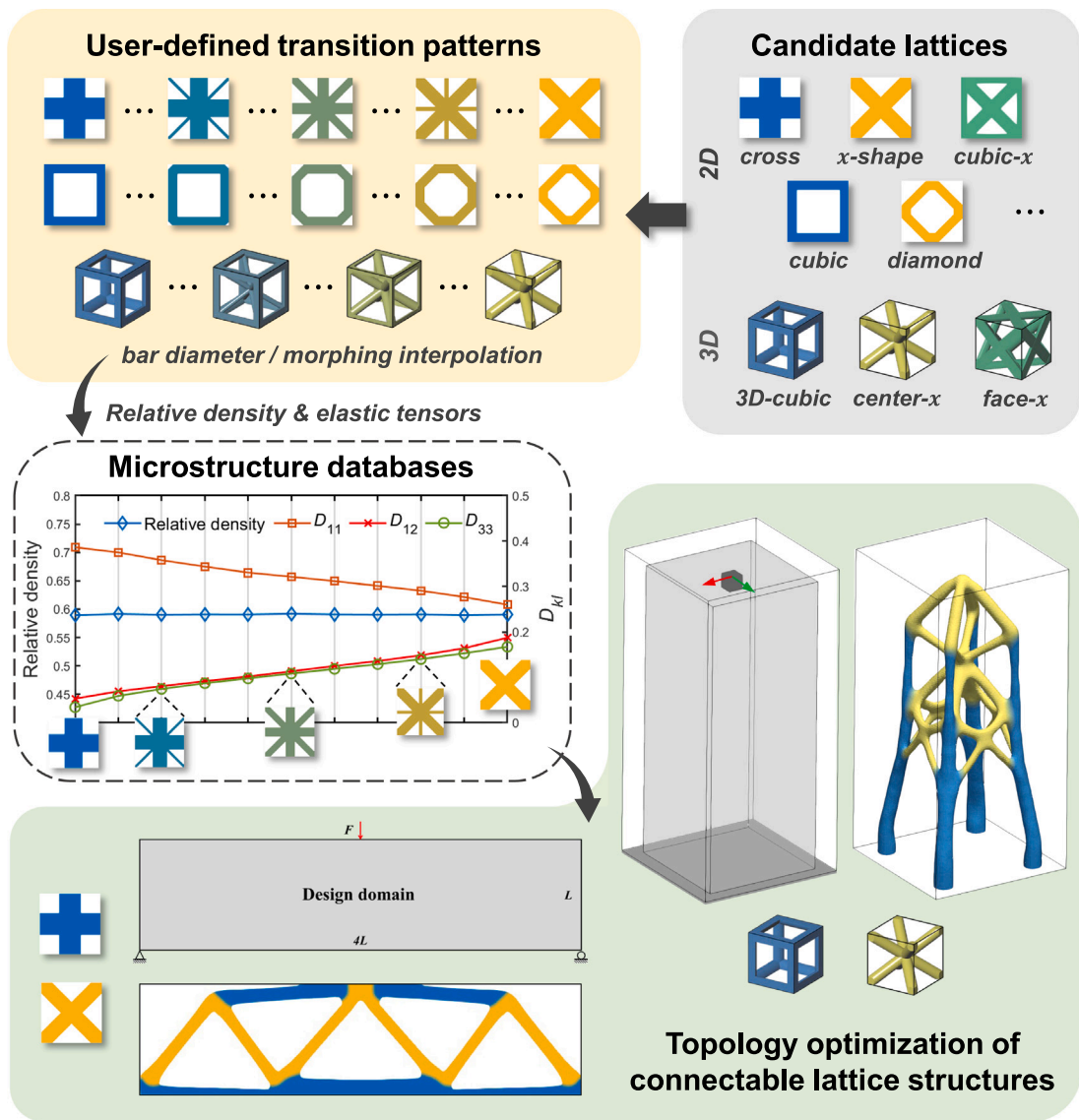


Fig. 2. Illustration of the proposed multi-domain topology optimization methodology for connectable lattice structures.

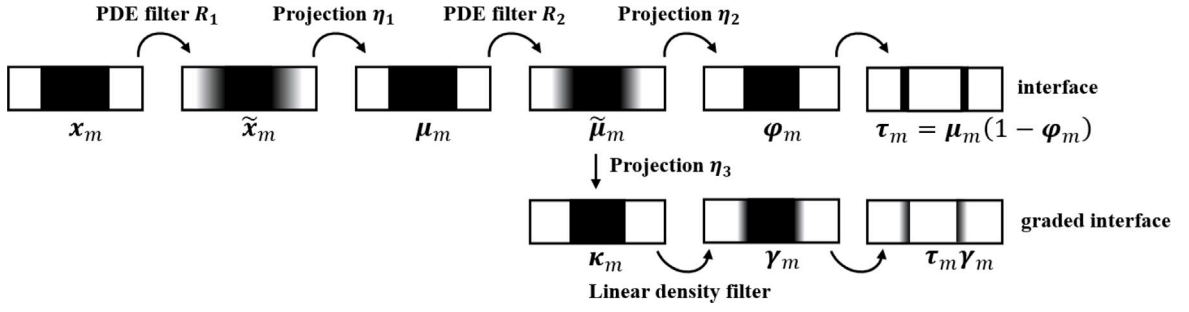


Fig. 3. Three-step filtering approach for describing the graded structural interface.

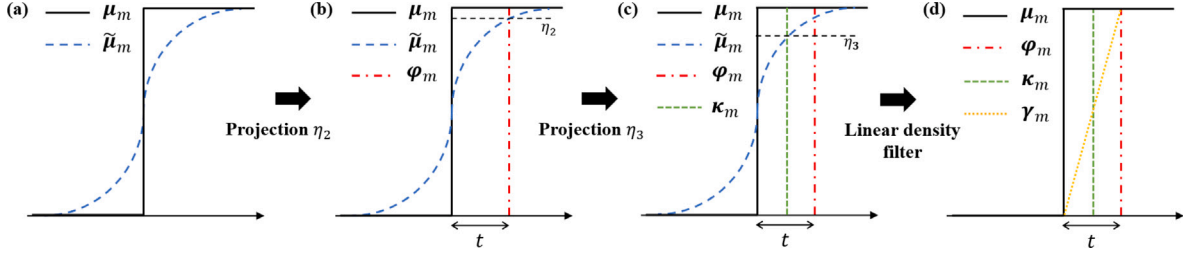


Fig. 4. 1D example of each field using three-step filtering approach.

2.2. Three-step graded interface identification method

The erosion-based interface identification method, which consists of two-step PDE-filtering and two-step projection, uses a relatively large threshold value to truncate the smoothed variable and the difference set of the original field μ_m and eroded field φ_m is identified as the structural interface, as shown in Fig. 3. The PDE filter [59] is based on a Helmholtz-type partial differential equation, which is suitable for parallel implementation. The filtered field $\tilde{\xi}$ is defined as a solution to the Helmholtz PDE applied to the initial field ξ :

$$-r^2 \nabla^2 \tilde{\xi} + \tilde{\xi} = \xi \quad (1)$$

where r is a length-scale parameter, which plays a similar role as the filter radius R in the standard filter. A relation between the length parameter r and the standard filter radius R can be expressed as:

$$r = \frac{R}{2\sqrt{3}} \quad (2)$$

To project intermediate values onto binary solutions, the continuous Heaviside function [60] is used:

$$\bar{\xi} = \frac{\tanh(\beta\eta) + \tanh(\beta(\tilde{\xi} - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))} \quad (3)$$

where β represents the sharpness of the Heaviside function, and $\eta \in [0, 1]$ is the threshold parameter.

In two-step projection operation, the first threshold η_1 is chosen as 0.5, and the second threshold η_2 is chosen as a value that larger than η_1 to obtain an erosion field. Then the structural interface is modeled by subtracting the eroded field μ_m from the intermediate field φ_m . For clarity, the threshold value η can be represented by a threshold bias $\Delta\eta \in (0, 0.5)$:

$$\eta = 0.5 + \Delta\eta \quad (4)$$

The interface thickness t depends on both PDE-filter radius of R_2 and threshold bias $\Delta\eta_2$, which can be formulated as

$$t = -\frac{R_2}{2\sqrt{3}} \ln(1 - 2\Delta\eta_2) \quad (5)$$

Details of analytical derivations for Eq. (5) can be found in [41]. Usually, the threshold value of the second Heaviside projection $\Delta\eta_2$ is chosen as 0.45, and the corresponding thickness of the interface t can be accurately controlled by

$$t \approx 0.67 R_2 \quad (6)$$

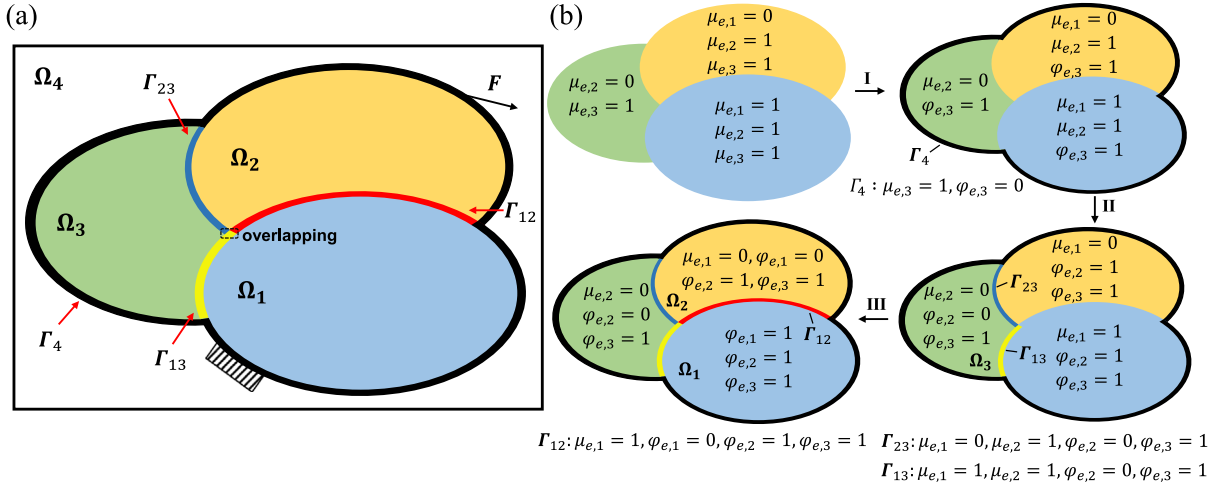


Fig. 5. Illustration of SIMP-based interpolation scheme of multi-domain connectable lattice structures: I, II and III represent the erosion and subtraction operations for μ_3 , μ_2 , and μ_1 , respectively.

As mentioned above, constructing a graded interface is the key that accurately arranging the intermediate unit cells at the specified transitional region. In this work, we extend two-step erosion-based method to generate a graded interface. Specifically, we perform Heaviside projection based on the smoothed field $\tilde{\mu}_m$ using an intermediate threshold value between η_1 and η_2 , aiming to erode the projected field κ_m from the original field μ_m by a distance of $t/2$. According to Eq. (5), it follows

$$t/2 = -\frac{R_2}{2\sqrt{3}} \ln(1 - 2\Delta\eta_3) \quad (7)$$

And we yield

$$\Delta\eta_3 = \frac{1}{2}(1 - \sqrt{1 - 2\Delta\eta_2}) \quad (8)$$

In this way, the boundary of κ_m is exactly the central axis of the interface τ_m . Then, we apply linear density filtering [61,62] to κ_m

$$\gamma_{e,m} = \sum_{i \in N_e} w_{e,i} \kappa_{i,m} \quad (9)$$

where $w_{e,i}$ are weighting factors based on the distance between element e and element i , and it follows $\sum_{i \in N_e} w_{e,i} = 1$ for $\forall e$. In this work, the constant weights, where all weight factors in the neighborhood are taken equal, are chosen to ensure filtered field is approaching a linear distribution:

$$w_{e,i} = \begin{cases} \frac{1}{|N_e|} & i \in N_e \\ 0 & i \notin N_e \end{cases} \quad (10)$$

where N_e represents the set of neighborhood elements in a square window of size $(t+1) \times (t+1)$, centered at element e . The linear density filtering with constant weights computes the average value of element e in the neighborhood. The field γ_m , a product of linear density filtering, acts as an indexing variable to determine the relative position of transition elements in the interface region and is monotonically distributed along the thickness direction. Taking one-dimensional design variables as an example, the fields mentioned above can be graphically represented in Fig. 4. As we can see, the field γ_m follows a 0–1 linear distribution along the thickness direction of the transition region, assuming that the threshold η_3 and window size are given according to the above formula.

2.3. Multi-domain interpolation scheme

In this subsection, a general recursively defined SIMP-based multi-domain interpolation formula considering piece interface is established. Assuming that microstructures within different domains are all not connected, we should separately design their transitional unit cells for each pair combination for building connectable microstructures over the whole structure. In addition to identifying all interfaces, each piece should be distinguished so that transitional unit cells attached to distinct matched pairs can be embedded correspondingly and accurately.

Taking four-domain as an example, each base phase (Ω_1 , Ω_2 , Ω_3 , and void phase Ω_4) and each interface layer between domain i and domain j (Γ_{12} , Γ_{13} , and Γ_{23}) should be separately defined, as illustrated in Fig. 5a. The classical SIMP-based multi-material

interpolated relative density ρ_e and elasticity matrix $\mathbf{D}_e$ of element e can be formulated as

$$\rho_e = \mu_{e,3}\mu_{e,2}\mu_{e,1}\rho^{(1)} + \mu_{e,3}\mu_{e,2}(1 - \mu_{e,1})\rho^{(2)} + \mu_{e,3}(1 - \mu_{e,2})\rho^{(3)} + (1 - \mu_{e,3})\rho^{(4)} \quad (11)$$

$$\mathbf{D}_e = \mu_{e,3}^p\mu_{e,2}^p\mu_{e,1}^p\mathbf{D}^{(1)} + \mu_{e,3}^p\mu_{e,2}^p(1 - \mu_{e,1}^p)\mathbf{D}^{(2)} + \mu_{e,3}^p(1 - \mu_{e,2}^p)\mathbf{D}^{(3)} + (1 - \mu_{e,3}^p)\mathbf{D}^{(4)} \quad (12)$$

in which $\rho^{(i)}$ and $\mathbf{D}^{(i)}$ ($i = 1, 2, 3$, and 4) represent the relative density and homogenized elasticity matrix of i th domain, respectively. p is the penalty factor. Usually, void phase Ω_4 is filled with weak materials to handle the numerical singularity issue.

For the convenience of readers to understand, we recursively perform the erosion and subtraction on each smoothed field μ_m , as shown in Fig. 5b. Firstly, we erode field μ_3 , i.e., the weight of solid material phases in the SIMP-based multi-domain method, and subtract its product φ_3 to obtain structural shell. If necessary, one can separately define the shell piece attached to different solid material phases based on fields μ_m . Then, we perform the erosion and extraction operations again on field μ_2 to identify the union interface layer $\Gamma_{13} \cup \Gamma_{23}$ and base on field μ_1 to further distinguish Γ_{13} and Γ_{23} , which can be represented as $\varphi_3(1 - \varphi_2)\mu_2\mu_1$ and $\varphi_3(1 - \varphi_2)\mu_2(1 - \mu_1)$, respectively. Analogously, interface layer Γ_{12} is derived following the operations mentioned above and represented as $\varphi_3\varphi_2(1 - \varphi_1)\mu_1$.

Note that multiple domains may share one interface layer, as shown in Fig. 5a. This overlapping region is quite complicated since it connects with three interface layers. In the above definition process, the overlapping region is prioritized as interface layer Γ_{13} , and this may lead to a disconnect. If this happens, one can additionally define a “mutual” microstructure. Correspondingly, the overlapping region can be represented as $\varphi_3(1 - \varphi_2)(1 - \varphi_1)\mu_2\mu_1$, and the interface layer Γ_{13} should be modified to $\varphi_3(1 - \varphi_2)\varphi_1\mu_2\mu_1$. This will be further discussed later.

Assume that we have $M + 1$ domains, the corresponding interpolated relative density and elasticity matrix can be expressed as

$$\rho_e = \sum_{i=1}^{M+1} \chi_{e,i} \rho^{(i)} + \sum_{i=1}^M \sum_{j=i+1}^{M+1} \chi_{e,ij} \rho_e^{(ij)} \quad (13)$$

$$\mathbf{D}_e = \sum_{i=1}^{M+1} \psi_{e,i} \mathbf{D}^{(i)} + \sum_{i=1}^M \sum_{j=i+1}^{M+1} \psi_{e,ij} \mathbf{D}_e^{(ij)} \quad (14)$$

with

$$\chi_{e,i} = (1 - \mu_{e,i-1})(1 - \varphi_{e,i-1}) \left(\prod_{j=i}^M \varphi_{e,j} \right) \quad (15)$$

$$\chi_{e,ij} = (1 - \mu_{e,i-1}) \left(\prod_{l=i}^{j-1} \mu_{e,l} \right) (1 - \varphi_{e,j-1}) \left(\prod_{l=j}^M \varphi_{e,l} \right) \quad (16)$$

$$\psi_{e,i} = (1 - \mu_{e,i-1})^p (1 - \varphi_{e,i-1})^p \left(\prod_{j=i}^M \varphi_{e,j}^p \right) \quad (17)$$

$$\psi_{e,ij} = (1 - \mu_{e,i-1})^p \left(\prod_{l=i}^{j-1} \mu_{e,l}^p \right) (1 - \varphi_{e,j-1})^p \left(\prod_{l=j}^M \varphi_{e,l}^p \right) \quad (18)$$

where $\rho_e^{(ij)}$ and $\mathbf{D}_e^{(ij)}$ represent the relative density and homogenized elasticity matrix of material in interface layer Γ_{ij} ($i < j$), respectively. The terms $\mu_{e,i-1}$ and $\varphi_{e,i-1}$ are set to 0 when $i = 1$. Due to the material properties at the interface being not constant but determined by the given transitional unit cells, $\rho_e^{(ij)}$ and $\mathbf{D}_e^{(ij)}$ are the functions of indexing quantity $\gamma_{e,j-1}$, and they can be obtained by interpolating based on microstructure offline databases. Note that we modify the penal strategy to ensure that each term is a convex expression [63]. Numerical experiments show that using a convex interpolation function can promote the structure to converge binary solutions.

Eqs. (13) and (14) describe a general situation that involves all underlying interface pieces. In this work, we mainly focus on the connectivity issue. The whole structure is composed of M solid material phases and one void phase Ω_{M+1} , and the structural shell is not considered here. Therefore, $\frac{1}{2}M(M - 1)$ types of transition patterns are specified to ensure each interface is well-connected. For convenience, just replace material in interface layers $\Gamma_{i,M+1}$ ($i = 1, \dots, M$) with weak material, and this will degrade to case of lattice-void structure without shell.

3. Topology optimization formulation

The optimization problem with the objective of compliance minimization under volume constraint can be mathematically written as

$$(\mathbb{P})_{nf}^1 = \begin{cases} \min_{\mathbf{x}_1, \dots, \mathbf{x}_M} & C = \mathbf{F}^T \mathbf{U}(\mathbf{x}_1, \dots, \mathbf{x}_M) \\ \text{s.t.} & v_i(\mathbf{x}_1, \dots, \mathbf{x}_M) - \bar{v}_i \leq 0, \quad i = 0, 1, \dots, M \\ & 0 \leq x_{e,m} \leq 1, \quad e = 1, 2, \dots, N, \quad m = 1, 2, \dots, M \\ \text{with} & v_0 = \sum_{e=1}^N \rho_e, \quad v_i = \sum_{e=1}^N \chi_{e,i} \rho^{(i)} \end{cases} \quad (19)$$

where C is the compliance, $\mathbf{F}$ and $\mathbf{U}$ are the external force vector and the displacement vector, respectively. $\mathbf{U}$ can be determined by solving the structural equilibrium equation $\mathbf{KU} = \mathbf{F}$ with $\mathbf{K}$. v is the material volume, and $\bar{v}$ represents the maximum volume

limit. Here, we define $M + 1$ volume constraints, i.e., the volume fraction for total materials $\bar{v}_0$ and for each material domain $\bar{v}_i$, $i = 1, \dots, M$, respectively. This simultaneously restrict the material at the interface. Other quantities have been defined above.

3.1. Sensitivity analysis

In this subsection, the sensitivity is analytically derived. According to the adjoint method, the derivatives of objective function C to the design variable $x_{e,m}$ can be written as

$$\frac{\partial C}{\partial x_{e,m}} = - \sum_{n=1}^N u_n^T \frac{\partial k_n}{\partial x_{e,m}} u_n = -u_n^T \left(\int_{\Omega_n} \mathbf{B}^T \frac{\partial \mathbf{D}_n}{\partial x_{e,m}} \mathbf{B} d\Omega_n \right) u_n \quad (20)$$

where u_n is the element displacement vector, and k_n is the element stiffness matrix. $\mathbf{B}$ is the strain–displacement matrix and Ω_n is the domain of the element. $\mathbf{D}_n$ is the elasticity matrix for n th element, which can be interpolated by Eq. (14).

Substituting Eq. (14) into $\partial \mathbf{D}_n / \partial x_{e,m}$, we yield

$$\frac{\partial \mathbf{D}_n}{\partial x_{e,m}} = \sum_{i=1}^{M+1} \frac{\partial \psi_{n,i}}{\partial x_{e,m}} \mathbf{D}_n^{(i)} + \sum_{i=1}^M \sum_{j=i+1}^{M+1} \frac{\partial \psi_{n,i,j} \mathbf{D}_n^{(ij)}}{\partial x_{e,m}} \quad (21)$$

To eliminate $\partial \psi_{n,i,j} \mathbf{D}_n^{(ij)} / \partial x_{e,m}$, the differentiation with respect to $x_{e,m}$ can be expressed as

$$\frac{\partial \psi_{n,i,j} \mathbf{D}_n^{(ij)}}{\partial x_{e,m}} = \frac{\partial \psi_{n,i,j}}{\partial x_{e,m}} \mathbf{D}_n^{(ij)} + \psi_{n,i,j} \frac{\partial \mathbf{D}_n^{(ij)}}{\partial x_{e,m}} \quad (22)$$

The term $\partial \mathbf{D}_n^{(ij)} / \partial x_{e,m}$ can be further derived as

$$\frac{\partial \mathbf{D}_n^{(ij)}}{\partial x_{e,m}} = \frac{\partial \mathbf{D}_n^{(ij)}}{\partial \gamma_{n,j-1}} \frac{\partial \gamma_{n,j-1}}{\partial x_{e,m}} \quad (23)$$

At the same time, the derivatives of volume constraints v_i to the design variable $x_{e,m}$ can be written as

$$\frac{\partial v_0}{\partial x_{e,m}} = \sum_{n=1}^N \frac{\partial \rho_n}{\partial x_{e,m}} \quad (24)$$

$$\frac{\partial v_i}{\partial x_{e,m}} = \sum_{n=1}^N \frac{\partial \chi_{n,i} \rho_n^{(i)}}{\partial x_{e,m}} \quad (25)$$

Substituting Eq. (13) into $\partial \rho_n / \partial x_{e,m}$, we yield

$$\frac{\partial \rho_n}{\partial x_{e,m}} = \sum_{i=1}^{M+1} \frac{\partial \chi_{n,i}}{\partial x_{e,m}} \rho_n^{(i)} + \sum_{i=1}^M \sum_{j=i+1}^{M+1} \frac{\partial \chi_{n,i,j} \rho_n^{(ij)}}{\partial x_{e,m}} \quad (26)$$

with

$$\frac{\partial \chi_{n,i,j} \rho_n^{(ij)}}{\partial x_{e,m}} = \frac{\partial \chi_{n,i,j}}{\partial x_{e,m}} \rho_n^{(ij)} + \chi_{n,i,j} \frac{\partial \rho_n^{(ij)}}{\partial x_{e,m}} \quad (27)$$

$$\frac{\partial \rho_n^{(ij)}}{\partial x_{e,m}} = \frac{\partial \rho_n^{(ij)}}{\partial \gamma_{n,j-1}} \frac{\partial \gamma_{n,j-1}}{\partial x_{e,m}} \quad (28)$$

The terms $\partial \mathbf{D}_n^{(ij)} / \partial \gamma_{n,j-1}$ and $\partial \rho_n^{(ij)} / \partial \gamma_{n,j-1}$ can be calculated by numerical derivative method from the microstructure database. And $\partial \gamma_{n,j-1} / \partial x_{e,m}$ can be written as

$$\frac{\partial \gamma_{n,j-1}}{\partial x_{e,m}} = \frac{\partial \gamma_{n,j-1}}{\partial \kappa_{l,j-1}} \frac{\partial \kappa_{l,j-1}}{\partial x_{e,m}} = \begin{cases} \sum_{l \in N_n} \frac{1}{|N_n|} \frac{\partial \kappa_{l,m}}{\partial x_{e,m}} & j-1 = m \\ 0 & j-1 \neq m \end{cases} \quad (29)$$

$\partial \varphi_{n,m} / \partial x_{e,m}$ and $\partial \mu_{n,m} / \partial x_{e,m}$ are further required to derive $\partial \psi_{n,i} / \partial x_{e,m}$, $\partial \psi_{n,i,j} / \partial x_{e,m}$, $\partial \chi_{n,i} / \partial x_{e,m}$ and $\partial \chi_{n,i,j} / \partial x_{e,m}$, according to Eqs. (13) and (14). Due to space limitations, the derivation of these intermediate terms is omitted here, but readers can refer to [41]. Moreover, $\partial \kappa_{l,m} / \partial x_{e,m}$ in Eq. (29) can be derived in the same way as $\partial \varphi_{n,m} / \partial x_{e,m}$. Finally, all the required terms for sensitivity analysis have been given above.

Based on the sensitivity analysis above, the moving asymptotes (MMA) solver is utilized to iteratively update the design variables [64]. For 3D problems, the topology optimization framework is based on the Portable and Extendable Toolkit for Scientific Computing (PETSc), referring to Aage's and Zhang's codes [65,66].

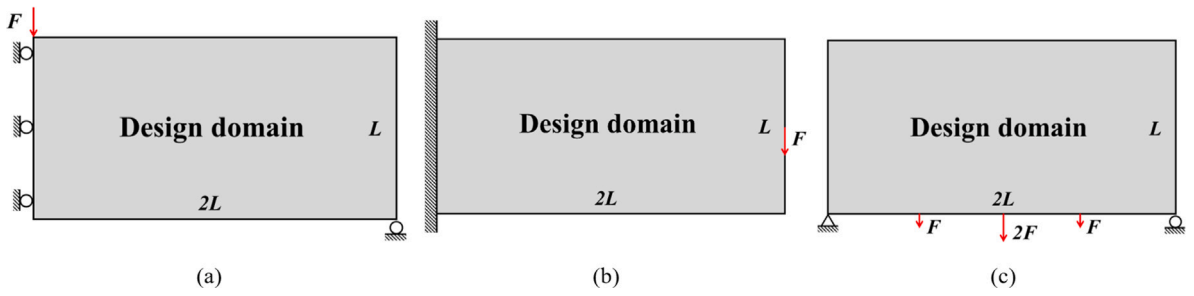


Fig. 6. The design domains and boundary conditions for 2D numerical examples: (a) MBB beam; (b) cantilever beam; (c) three-point beam.

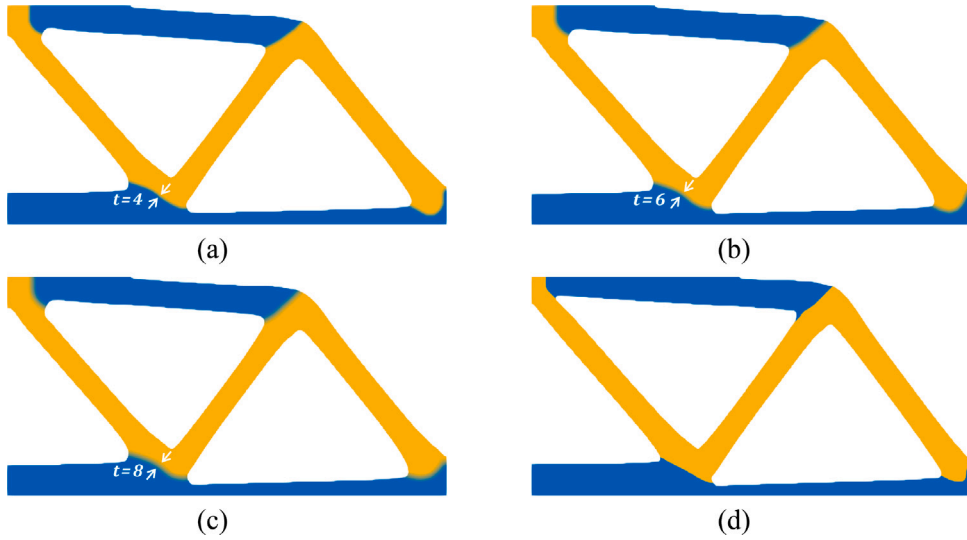


Fig. 7. Optimized MBB beam with varying interface thickness: (a) $t = 4$, $C = 337.02$; (b) $t = 6$, $C = 339.72$; (c) $t = 8$, $C = 342.04$; (d) classic multi-material design without considering transitional unit cells, $C = 336.69$.

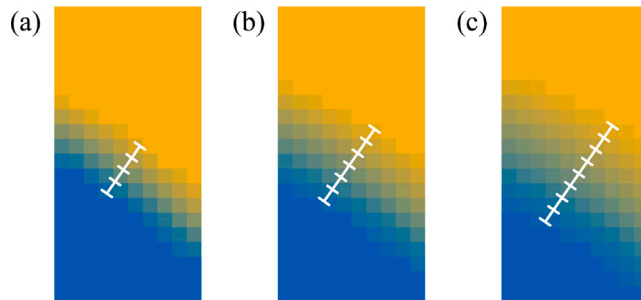


Fig. 8. Details of optimized MBB beam with varying interface thickness: (a) $t = 4$; (b) $t = 6$; (c) $t = 8$. Scale bar indicates 1.0.

4. Numerical verification

In this section, several 2D and 3D numerical examples are employed to verify the stability, effectiveness, and scalability of the proposed method. The design domains and boundary conditions are presented in Fig. 6 for 2D numerical examples. Unless otherwise stated, the Young's modulus $E_s = 1$ for solid material, $E_v = 10^{-9}$ for weak material, and Poisson's ratio $\nu = 0.3$ for all tests. The concentrated force is set to $F = 1$. The convergence condition is set when each change in 10 iterations is less than 0.1%. A continuation scheme is adopted, where β is gradually increased from 1 to 128 by doubling at every 50 iterations (or at convergence).

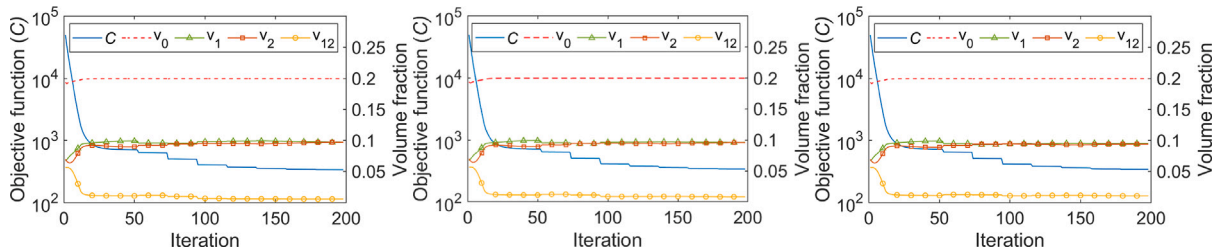


Fig. 9. The iteration histories for optimized structures: $t = 4$ (left), $t = 6$ (middle), and $t = 8$ (right).

4.1. MBB beam

In the first example, we aim to verify the controllability of the graded interface and the convergence behavior of the proposed algorithm. The candidate lattices are chosen as cross and x-shape microstructures with a bar diameter of $d = 0.36$. We collect the relative density and homogenized elasticity tensors of 9 transitional unit cells, which are constructed by bar diameter interpolation, as shown in Fig. 2. The design domain of MBB beam is discretized into 400×200 quadrilateral elements, and the dimension is set to $L = 200$. The total volume fraction $\bar{v}_0 = 0.2$ and the respective volume fraction for each candidate lattice $\bar{v}_1 = \bar{v}_2 = 0.1$. The first filter radius R_1 is set to 20. We consider three different interface thicknesses, i.e., $t = 4, 6$, and 8. According to Eq. (6), the filter radius R_2 is determined as 6, 9, and 12. Here, the classic multi-material design in which the transitional unit cells are ignored is also provided.

The optimized lattice structures are shown in Fig. 7, in which the cross and x-shape microstructures are expressed in blue and yellow, respectively. The interface layer between two domains is highlighted with an RGB-based interpolated color map according to the field γ_{j-1} , which indicates the graded variation of transitional unit cells. From the optimized structures, the cross microstructures that perform better in the horizontal direction are distributed in the tensile and compressive regions, while the x-shape microstructures tend to be distributed in the regions of inclined bars due to superior shear resistance. Compared with traditional multi-material design, as shown in Fig. 7d, the interface layer obtained by the proposed method is smoother due to considering continuous and varying transitional unit cells. Further, from the amplified details provided in Fig. 8, the interface thickness and graded varying features are accurately defined in all cases.

The iteration histories of three cases are given in Fig. 9. Three volume fraction constraints are all stably satisfied during the optimization process. The volume fractions v_1 and v_2 are not strictly approaching the limit value due to the transitional unit cells being involved in the total constraint $\bar{v}_0$. The volume fractions of transitional unit cells in three cases are 0.57%, 0.82%, and 1.08%, respectively. This trend is reasonable because the interface thickness is increasing. Note that the abrupt change points in the convergence curves are caused by doubling the sharpness parameter β . Assuming that reference designs are eroded and subtracted operations to the topology shown in Fig. 7d, their equivalent compliance is 338.2, 340.62, and 343.85 for $t = 4, 6$, and 8, respectively. It is evident that the equivalent compliance is all higher than the optimal solution provided in Fig. 7a-c, which shows the structures obtained by the proposed method with consideration of transitional unit cells are better than traditional multi-material design.

4.2. Cantilever beam

An optimized cantilever beam is presented here to investigate the potential impact of the varying patterns of transitional unit cells. The candidate lattices are chosen as cubic and diamond microstructures with a bar diameter of $d = 0.3$. The design domain of cantilever beam is discretized into 100×50 quadrilateral elements, and the dimension is set to $L = 50$. The total volume fraction $\bar{v}_0 = 0.2$ and the respective volume fraction for each candidate lattice $\bar{v}_1 = \bar{v}_2 = 0.1$. The first filter radius R_1 is set to 5. The interface width $t = 2$. According to Eq. (6), the filter radius R_2 is determined to be 3. Three kinds of transitional patterns are considered. In case 1, we strictly interpolate based on bar diameter among the base lattice and transitional unit cells. Different from case 1, we enforce a minimum bar diameter limit $d_{\min} = 0.1$ for the first and last transitional unit cells in case 2, aiming to satisfy manufacturing resolution. Therefore, a mutation can be observed in the curve of case 2 shown in Fig. 10(b). In case 3, we interpolate based on shape-morphing and ensure the bar diameter of each microstructure is constant.

The reconstructed full-scale structures are provided in Fig. 10. The objective function in three cases is 304.77, 302.23, and 316.96, respectively. The structure obtained by shape interpolation tends to have degraded mechanical performance relative to that obtained by bar diameter interpolation, which may be attributed to the shape-morphing transitional unit cells with high bulk modulus but low stiffness [46]. From the amplified details, we can find that case 2 has larger bar diameters for transition unit cells compared with case 1, which is facilitated by the minimum bar diameter implementation and may further satisfy manufacturing resolution.

Ideally, the transitional unit cells should be exactly the linear interpolated configuration of $\gamma = 1/3$ and $\gamma = 2/3$ if the interface thickness is set to 2, according to the one-dimensional example shown in Fig. 4. However, when the thickness direction of the interface layer is not strictly parallel to the coordinate axis direction, the value of γ is different from the ideal one, further leading to the modeled transition unit cells deviating from the linear interpolated configuration. But it still follows the prescribed transitional trend. This phenomenon will be alleviated with an increased grid discretization of design domain and a large interface width.

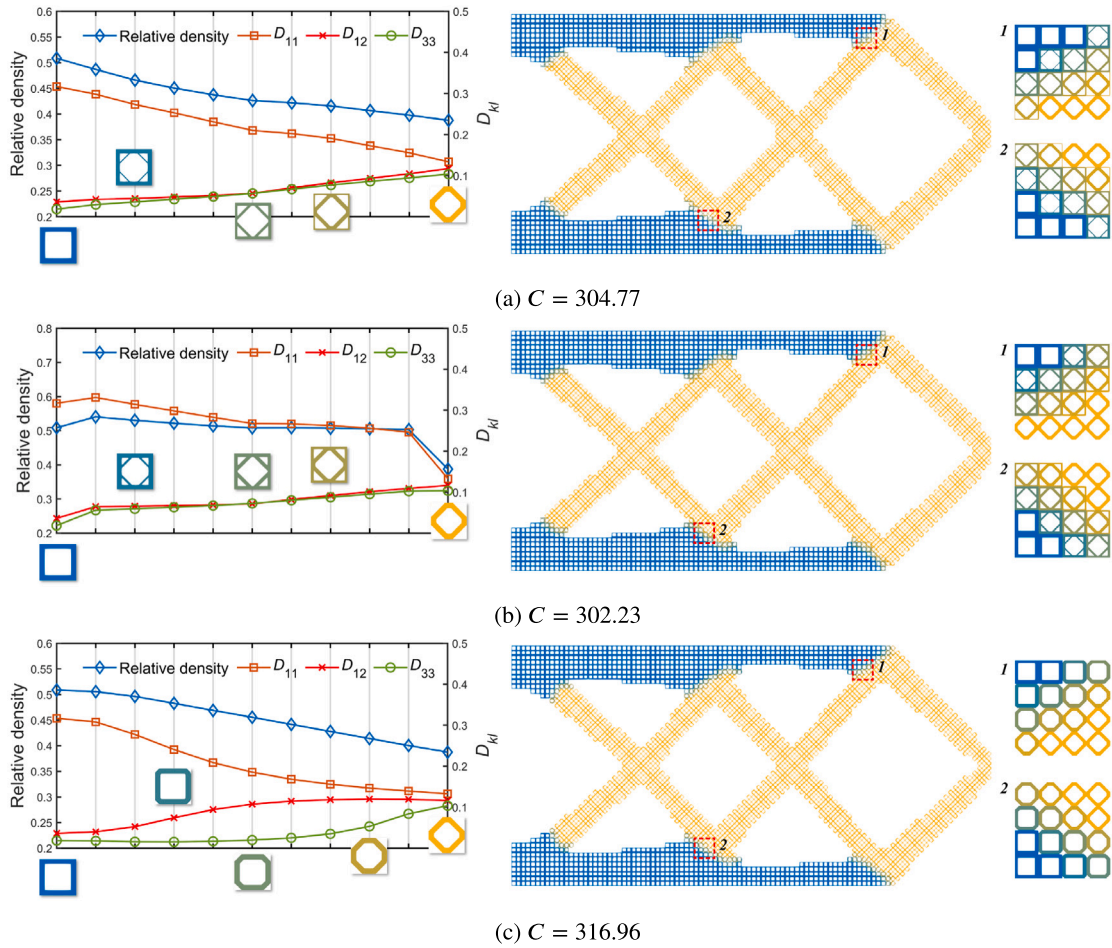


Fig. 10. Reconstructed full-scale cantilever beam with varying transition pattern.

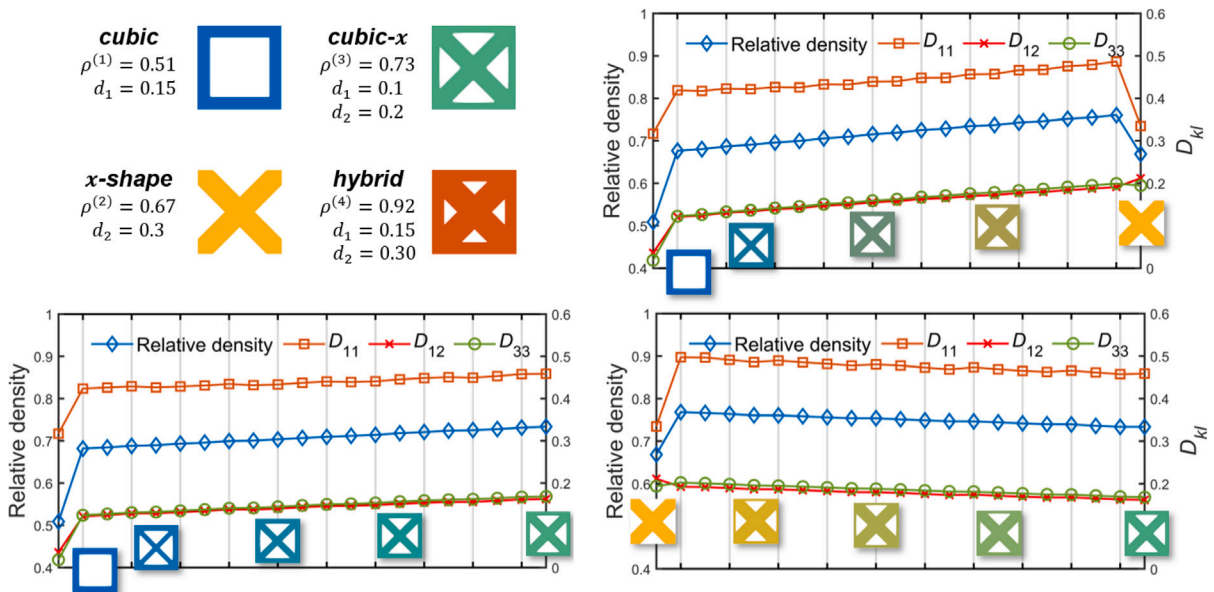


Fig. 11. Configurations of candidate microstructures and databases of transitional unit cells.

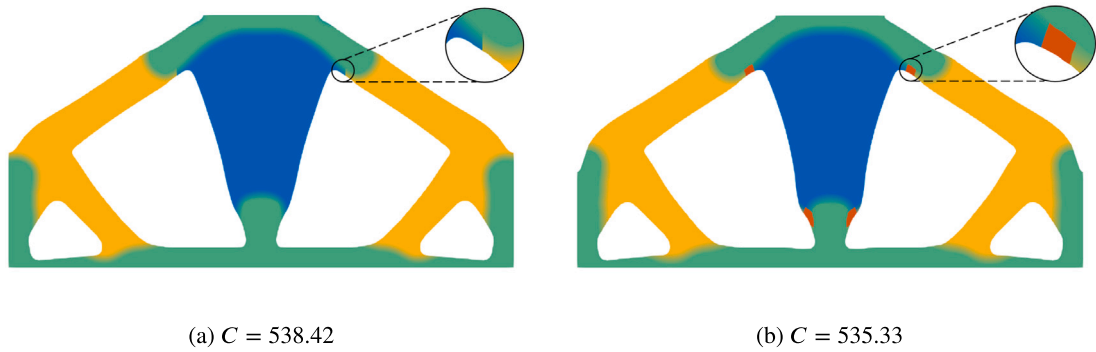


Fig. 12. Optimized structures considering three candidate microstructures: the overlapping region defined as Γ_{12} for case 1 and “mutual” microstructure for case 2.

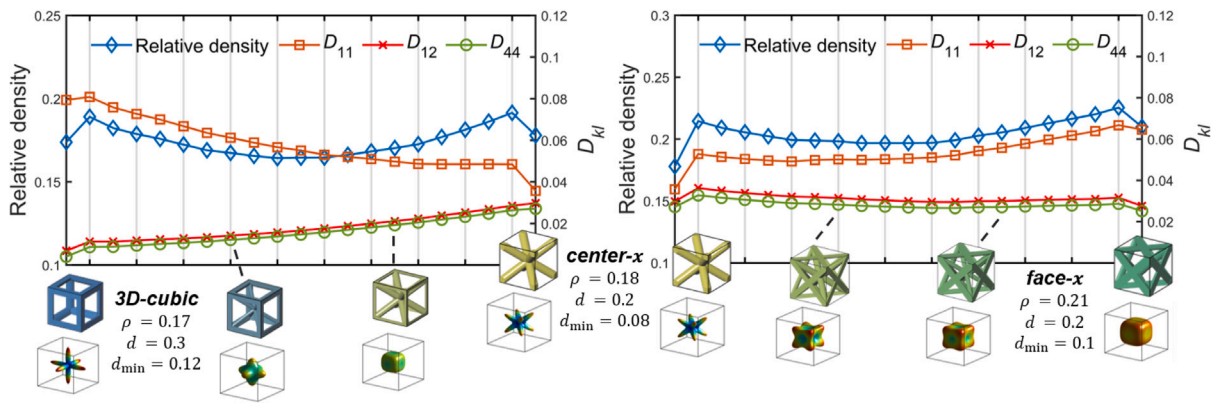


Fig. 13. Configurations of truss candidate microstructures and databases of transitional unit cells for 3D numerical examples.

4.3. Three-point beam

An optimized three-point beam is presented to demonstrate the ability to identify the interface layer separately for more than two lattice domains. The cubic, x -shape, and cubic- x lattices are chosen as the candidate microstructures, and their configurations and bar dimensions are shown in Fig. 11. The minimum bar diameter limits are enforced with $d_{\min} = 0.05$ for cubic lattice and $d_{\min} = 0.1$ for x -shape lattice. The design domain of three-point beam is discretized into 400×200 quadrilateral elements, and the dimension is set to $L = 200$. The total volume fraction $\bar{v}_0 = 0.3$ and their respective volume fraction for each candidate lattice $\bar{v}_1 = \bar{v}_2 = \bar{v}_3 = 0.1$. The first filter radius R_1 is set to 20. The interface width $t = 8$. Three transition patterns are considered to generate connectable microstructures over the whole structure. Note that although cubic and cubic- x lattices are well-connected, their transitional unit cells are defined to promote smoother continuity. For each transition pattern, 19 transitional unit cells are collected through bar diameter interpolation.

The optimized three-point beam is shown in Fig. 12(a), in which the cubic, x -shape, and cubic- x microstructures are expressed in blue, yellow, and green, respectively. Their separate interface layer Γ_{ij} is highlighted with an RGB-based interpolated color map according to field γ_{j-1} . For example, Γ_{12} is visualized based on γ_1 using interpolated colors between blue and yellow. From the optimized lattice structure, each piece of interface attached to different base microstructures is well-defined, which demonstrates the proposed algorithm is capable of distinguishing interface layers separately.

As discussed in Section 2.3, an overlapping region may grow, and it will be described as interface layer Γ_{13} first, as shown in the amplified detail of Fig. 12(a). Therefore, we define a “mutual” hybrid microstructure (colored orange) and modify the interpolation scheme according to Section 2.3. From the optimized result (Fig. 12(b)), we can find that the overlapping region can also be identified accurately. In addition, complex stress conditions in the overlapping region make microstructures that are close to isotropic more competitive. Therefore, the second case, where the “mutual” hybrid microstructure is closer to solid materials, tends to exhibit greater stiffness. But at the same time, due to the small overlapping region resulting in a low contribution to the overall stiffness, the objective functions of the two cases will not differ significantly.

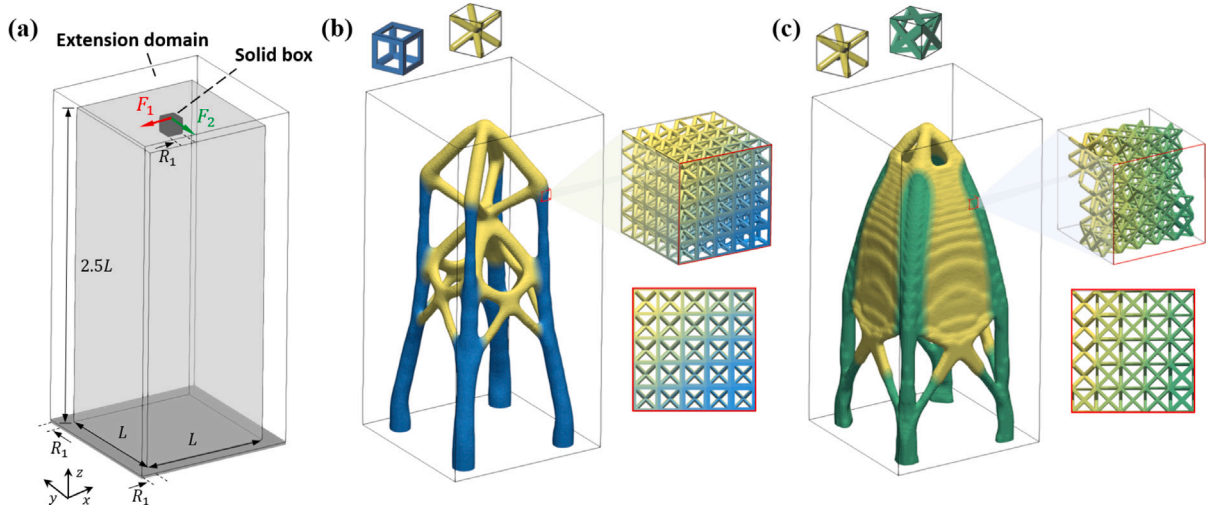


Fig. 14. (a) The design domain and boundary condition; (b) optimized 3D cantilever considering 3D-cubic and center-x microstructures, $C = 907$; (c) optimized 3D cantilever considering center-x and face-x microstructures, $C = 1067$.

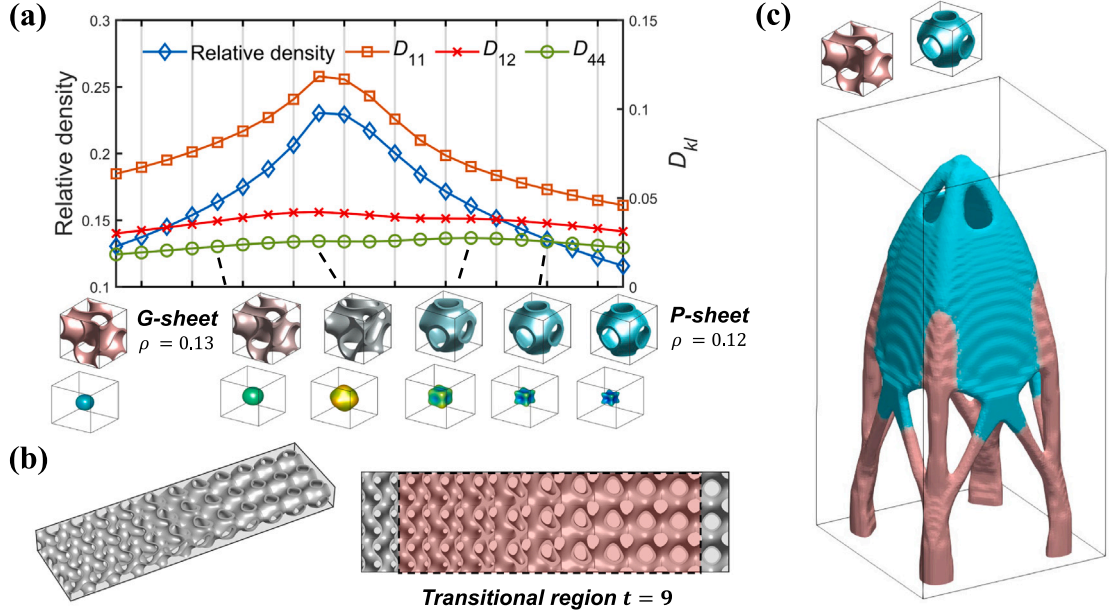


Fig. 15. (a) Configurations of TPMS-based candidate microstructures and databases of transitional unit cells; (b) a repeat pattern illustration with 9 transitional unit cells; (c) optimized 3D cantilever considering G-sheet and P-sheet microstructures, $C = 640$.

4.4. 3D cantilever

In this example, a design problem for a 3D cantilever structure is presented. The boundary conditions are given in Fig. 14a, where the dimension is set to $L = 64$. Two load cases are considered: a concentrated force $F = 1$ is applied at the center of the top surface along the x and y directions, respectively. In order to prevent materials from “sticking” to the domain boundaries, the extension approach is applied [67]. Additionally, a solid box is also used to remove the large, local deformations that would appear near the load region [54]. Note that the extension distance and the width of the solid box should be larger than the filter radius R_1 . The design domain is discretized by $64 \times 64 \times 160$ 8-node hexahedron elements. The total volume fraction $\bar{v}_0 = 0.01$ and the respective volume fraction for each candidate lattice $\bar{v}_1 = \bar{v}_2 = 0.5\%$. The first filter radius R_1 is set to 8. Therefore, the extended domain is discretized by $80 \times 80 \times 168$ elements for the FEM analysis. The interface width is set to $t = 4$. Three truss candidate microstructures are considered, i.e., 3D-cubic, center-x, and face-x, whose dimensions are illustrated in Fig. 13. For each transition pattern, 19 transitional unit cells are collected through bar diameter interpolation. As in Section 4.2, a minimum bar diameter limit

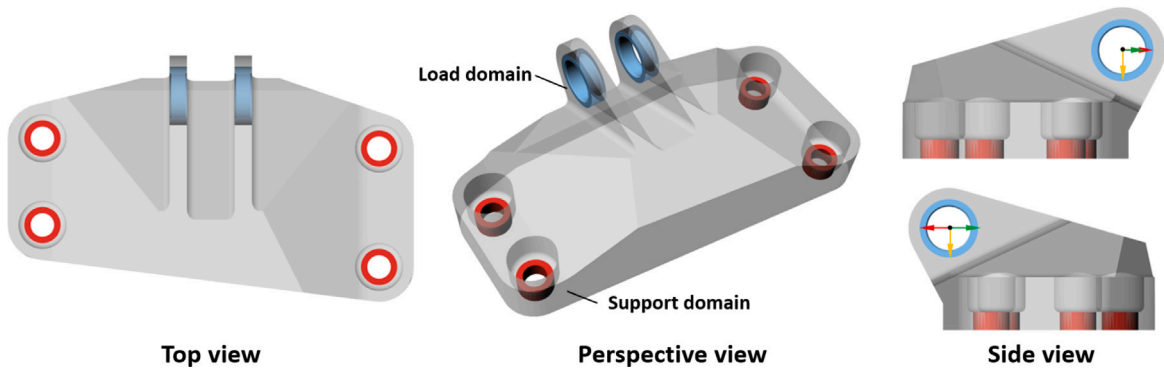
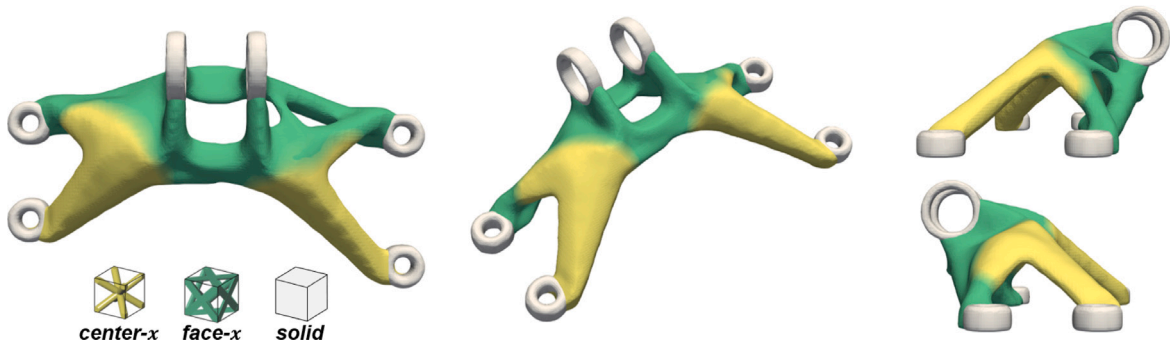


Fig. 16. The design domain and boundary condition of JEB.

Fig. 17. Optimized JEB with different views, $C = 7952$.

is set, aiming to satisfy manufacturing resolution.

We also consider two types of TPMS-based structures, i.e., G-sheet and P-sheet, whose construction process can refer to [44]. We collect 19 transitional unit cells, and each lattice can be derived with a parameterized implicit expression

$$\phi_{hyb}^i = (1 - \alpha)\phi_G + \alpha\phi_P, \quad i = 1, \dots, I \quad (30)$$

where ϕ_G , ϕ_P , and ϕ_{hyb}^i indicate the implicit functions used to define base lattices G-sheet, P-sheet, and i th transitional lattice, respectively. I is the number of collected transitional lattices. The expression α is a weight function that satisfies $\alpha = i/(I + 1)$. In addition, we also present a repeat pattern with 9 transitional unit cells in Fig. 15b, which shows the constructed transition unit cells can achieve a smooth connection between two different parent morphologies.

The optimized 3D cantilever structures considering different truss microstructures are shown in Fig. 14. Due to the large amount of full-scale structural data, we only reconstruct the local region highlighted with a red box. The optimized 3D cantilever structure considering TPMS-based microstructures are shown in Fig. 15c. All the optimized results become the structure frame with four supports far from the neutral axis of the beam, which is helpful in resisting bending moments from two load cases. From the Young's modulus provided in Figs. 13 and 15a, the microstructures with higher tensile stiffness tend toward the tension/compression regions, and materials with superior shear resistance tend toward the inclined members and regions of high shear. At the same time, all the graded interface layers are well-defined.

4.5. Jet engine bracket

The jet engine bracket (JEB) with 3D irregular design domain topology optimization problem is tested here. As shown in Fig. 16, the design domain of JEB is highlighted with gray color. The support domain and load domain are marked with red and blue, respectively, which are all set to passive solid region. Three load conditions are considered: vertical load (highlighted with yellow), horizontal load (highlighted with red), and torsional load (highlighted with green). For all load conditions, a distributed force $F = 1$ is applied at the load domain, and the inner surfaces of the four rings at the bottom of the JEB are fixed in all three directions. Each load weight is set equal in the objective function. The design domain is discretized by $88 \times 144 \times 48$ 8-node hexahedron elements. The total volume fraction $\bar{v}_0 = 0.02$ and the respective volume fraction for each candidate lattice $\bar{v}_1 = \bar{v}_2 = 0.01$. The first filter radius R_1 is set to 8, and the interface width is set to $t = 4$. Center- x and face- x lattices are considered, whose dimensions are illustrated in Fig. 13. The optimized JEB with different views is shown in Fig. 17. Results are written to vtu files for visualization using ParaView,

and a smoothing step is added (ParaView filter: Smooth) to generate a smoother surface. From the optimized structure, the center- x lattices tend to distribute in the truss-like legs of the structure, while the face- x lattices closer to isotropic are presented in regions that may be subjected to loading in multiple directions, which exhibits reasonable distribution and is consistent with [68]. In addition, the graded interface layer is well-defined, which showcases the scalability of the proposed method for irregular design domains.

5. Conclusions

In this work, a novel multi-domain topology optimization approach is proposed to handle connectivity issues between dissimilar microstructures. First, a set of transitional unit cells that follow a particular transition pattern is introduced to achieve multi-class but well-connected lattice structures. Such smooth and tunable transition patterns are realized by interpolation of bar diameter, shape-morphing, and parameterized implicit functions, which exhibit superior adaptability to disconnected microstructures with intricate geometric configurations. Next, the erosion-based interface identification method is extended to the graded interface model based on a linear density filter, which enables us to precisely embed the varying transitional unit cells at the specified interface. Additionally, a multi-domain interpolation scheme considering both base microstructures and transitional unit cells is proposed, in which each piece of interface layer can be further separately defined to navigate situations with more than two lattice material phases. Several 2D and 3D numerical examples are presented to demonstrate the stability, effectiveness, and scalability of the proposed method.

The proposed method exhibits superior adaptability when addressing complex connectivity problems, especially for lattices featuring complex configurations and conditions with multiple dissimilar base microstructures, by effective interpolation techniques or robust parameterization methods. In future work, the proposed method will be extended to functionally graded cellular structure optimization or concurrent optimization framework. In addition, the transitional unit cells will be generated by simultaneously optimizing, further enlarging the design space of multi-morphology microstructures and exploring the structural potential.

CRedit authorship contribution statement

Peng Wei: Writing – review & editing, Supervision, Methodology, Conceptualization. **Xinglong Chen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation. **Hui Liu:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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